Pro-Cache-CIM: A 28nm 69.4TOPS/W Product-Cache-based Digital-Compute-in-Memory Macro Leveraging Data Locality Pattern in Vision AI tasks

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Digital-based compute-in-memory macros (DCIM) and digital accelerators that leverage lookup tables (LUT)[1, 2, 3] have demonstrated remarkable advantages in achieving high energy efficiency while maintaining full precision. By pre-computing summation results, LUT-based DCIMs eliminate the energy consumption of one or two stages of addertrees. However, scaling these LUT-based DCIMs to higher orders—which requires using wider bit-widths for indexing LUT content—poses significant challenges. This is primarily due to the exponential growth in the combinations of pre-computed results, which consequently leads to proportional increases in area and power consumption, ultimately degrading both area and energy efficiency.

This work leverages the data locality pattern observed in vision AI tasks to overcome the scaling limitations of LUT-based DCIMs. In such tasks, the activations (ACT) within small regions tend to fall within a narrow range[4]. With the weight-stationary dataflow of typical CIMs, we can replace the large and inefficient LUT memory with a small, efficient cache that stores the products of ACTs and WTs. However, applying the product-cache to DCIMs presents several challenges: (1) The multipliers needed for handling cache misses result in significant power and area overhead; (2) The inherent high parallelism of DCIMs leads to a considerable stall rate, as the cache miss of any single channel can cause the entire CIM macro to stall.

To tackle these challenges, this paper introduces a product-cache-based DCIM macro featuring three key components: (1) A product-cache-based multiplier that leverages the data locality of ACTs and the stationary property of WTs to cache their products, achieving enhanced energy efficiency; (2) Inspired by [5, 6], a shared single-cycle booth16 multiplier that replaces 32 booth4 multipliers with 8 1W2R-6T-SRAM-based booth16 multipliers, significantly reducing power and area overhead; (3) A PRE^3 cache controller (preload, pre-lookup, pre-update) that anticipates future misses and preemptively updates the missed products, leading to improved overall throughput by minimizing stall cycles.

Fig.2 shows the overall architecture of the Product-Cache CIM macro, consisting of a PRE^3 cache controller, 16 product-cache-based multiplier banks, 16 addertrees, a top controller, and the buffers for ACT, PSUM, and pre-compute WTs. Each multiplier bank contains 32 Product-Caches and 8 single-cycle booth16 multipliers. Each Product-Cache features an 18x16 SRAM array, with each row corresponding to a cached product value. To perform the MAC operation, the PRE^3 cache controller first looks up the ACT value in the cache tag. If a cache miss occurs, the missed ACT is sent to the cache replacer and the multiplier to generate the replacement address and the missed product respectively. The address and the product are then sent to the Product-Cache to update it. Conversely, if a cache hit occurs, the PRE^3 controller transmits the hit address to the Product-Cache, which then reads out the cached product and sends it to the addertree. Once the products of all input channels are ready, the addertree completes the computation and stores the partial sum (PSUM) into the buffer.

Fig.3 shows the circuit-level details of the Product-Cache and the single-cycle booth16 multiplier (SCB16-MUL). The 1W2R 6T-SRAM cell[5] is utilized in the Product-Cache to reduce readout power, thanks to its precharge-free design and the low voltage swing of the bitline. An inverter is incorporated at the end of BLB to recover the threshold voltage loss. To reduce the power consumption of calculating the missed product, a booth16 multiplier[6] is employed in place of the conventional booth4 multiplier. The basic concept is constructing 0, $\pm$W, $\pm$2W, ..., $\pm$8W by shifting and negating the pre-computed 1/3/5/7W. However, the two-cycle latency of [6] results in additional stall cycles, negatively impacting overall throughput. To address this issue, the SCB16-MUL implements the 1/3/5/7W LUT (W-LUT) with a 1W2R 6T-SRAM array[5], thereby eliminating the necessity to duplicate the W-LUT and resulting in enhanced area efficiency. In comparison to the conventional booth4 multiplier, the Product-Cache achieves 4.3x power saving and 1.3x area saving; the SCB16-MUL delivers 1.5x power saving and 1.1x area saving. A SCB16-MUL is shared by four input channels to optimize area efficiency and multiplier utilization. Increasing the share ratio further leads to reduced area savings while significantly increasing the STALL rate.

Fig.4 shows the scheme of PRE^3 cache controller. The conventional cache controller handles unpredictable dataflows and only buffers the ACTs of the current cycles. When there are no misses, the shared multiplier is idle, but when multiple misses occur, it processes them serially, causing stalls of consecutive cycles. The proposed PRE^3 cache controller introduces three features: preload, pre-lookup, and pre-update. During preload, the value ACT[N+2], needed in two cycles, is added to the preload fifo. In pre-lookup, the controller looks up ACT[N+2] in the cache. If a cache miss occurs, the pre-update stage is activated, where the missed product (ACT[N+2]xW) is calculated with a SCB16-MUL and stored in the Product-Cache based on the replacement address. If pre-update requests of multiple channels are sent to the same multiplier, they are buffered and processed sequentially. The PRE^3 controller minimizes stalls and achieves a 1.9x throughput improvement. The number of cache lines and replacement strategy are chosen based on design space exploration results. A cache line count of 16 is selected to reduce the miss rate without significantly increasing area and power. The direct-mapped replacement strategy is adopted for its lowest miss rate and area cost. Implementation is slightly complex since when updating ACT[N+2]xW into the Product-Cache, ACT[N+1]xW and ACT[N]xW are not yet sent to the addertree and should not be replaced. To address this, 2 additional cache lines are added as buffer lines when there occurs the conflict of saving any two of ACT[N]/ACT[N+1]/ACT[N+2] to the same line.

Fig.5 shows the detailed chip measurement results. There are 5 power domains (PRE^3, WL/CL driver, Product-Cache, multiplier, and addertree) enabling fine-grain DVFS and breakdown measurement. The power supply voltage for the addertree is denoted as VDD_{AT}, while the voltage for the other power domains is denoted as VDD_{MAIN}. Given the significant leakage power of SRAM-based Product-Cache, VDD_{MAIN} is fixed at 0.8V to prevent frequency degradation at lower VDD_{AT}, thereby enhancing energy efficiency. The peak energy efficiency of 69.4 TOPS/W is achieved at VDD_{MAIN} of 0.8V and VDD_{AT} of 0.53V during the processing of layer 50 of ResNet-50 on the ImageNet dataset. Due to the limited area budget of the prototype chip, the power consumption is dominated by the PRE^3 controller. This is because the power consumption can be amortized over more output channels, achieving better energy efficiency. A balanced power breakdown is observed when the number of output channels is increased from 16 to 64.

Fig.6 shows the layer-wise measurement results of ResNet-50 and the comparison table. This work leverages the data locality patterns of vision AI tasks to develop a product-cache-based digital CIM macro. Unlike previous non-full-precision CIMs that sacrifice accuracy for reduced power consumption, this work maintains full precision while achieving comparable energy efficiency. In comparison to earlier full-precision DCIMs, this work shows an 1.5-3.8x energy efficiency improvement. Furthermore, energy efficiency can be enhanced by increasing the number of output channels to 64, yielding an 1.7-3.8x improvement over prior non-full-precision CIMs.

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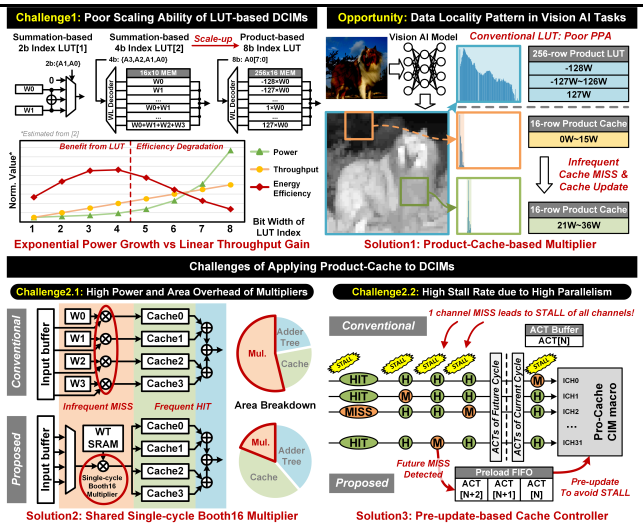


Fig. 1. Motivation and challenges of Product-Cache DCIM.

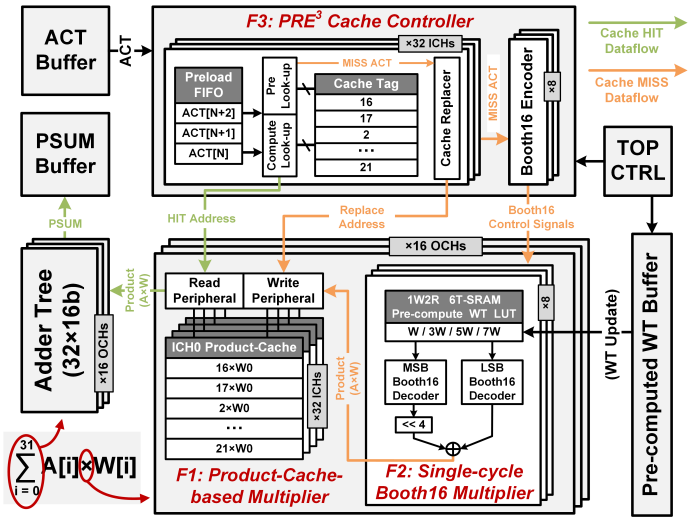


Fig. 2. The overall architecture and key features of this work.

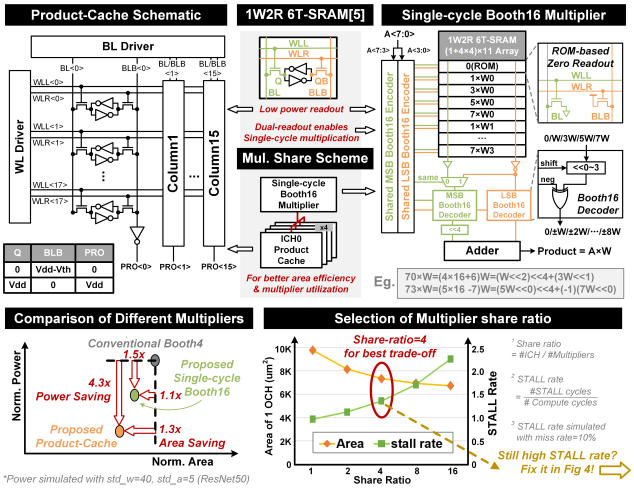


Fig. 3. Circuit-level details of Product-Cache and single-cycle booth16 multiplier; and the design trade-off of multiplier share ratio.

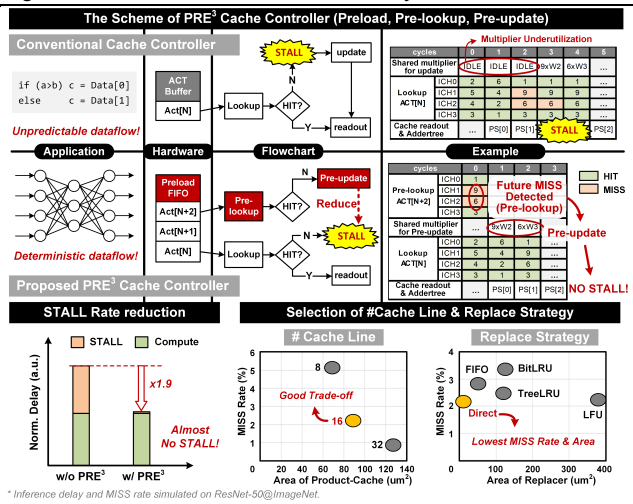


Fig. 4. The scheme and improvement of PRE^3 cache controller; and design choices of #cache-line and replace strategy.

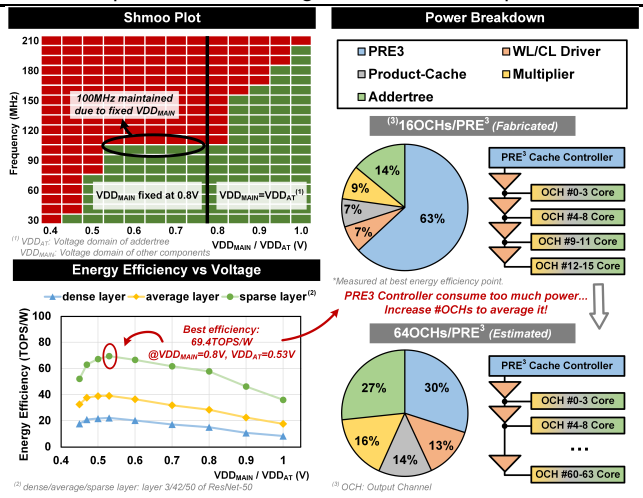
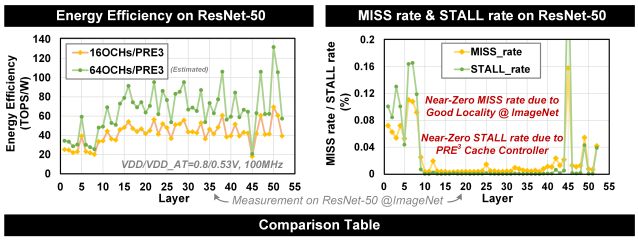


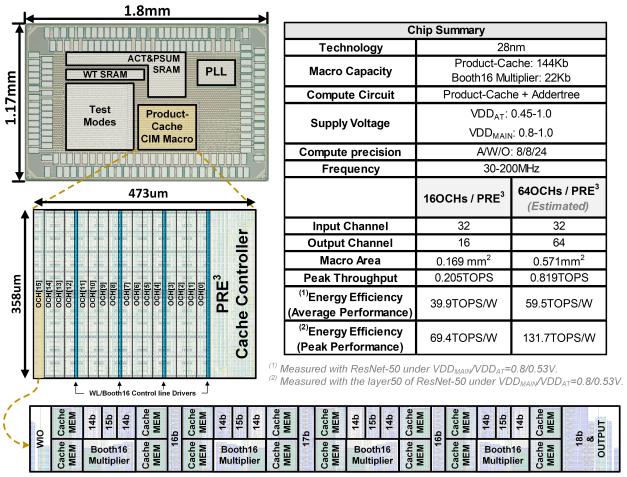
Fig. 5. Chip measurement results: Shmoo plot, voltage scaling vs energy efficiency and power breakdown.



	VLSI23 E. Chang	VLSI22 C2.5 C. Lee	ISSCC24 Y. He	This work	ISSCC24 A. Guo	CICC23 J. Song	ISSCC22 D. Wang
Technology	12nm	12nm	28nm	28nm	22nm	65nm	28nm
MAC Operation	Digital Addertree	Summation-LUT + Addertree	Summation-LUT + Addertree	Product-Cache + Addertree	Approx Digital +Current	Current-mode ACIM	Approx Digital
#Combination of Pre-Computation	\	4	16	256	\	\	\
Supply Voltage (V)	0.55-0.99	0.72-0.92	0.9-1.2	VDD_{AT}: 0.45-1.0 VDD_{MAIN}: 0.8-1.0	0.6-0.9	0.7-1.0	0.45-1.10
# of In/Out Ch	64/16	64/16	64/16	32/16	128/4	128/16	256/64
Bit Precision (bit)	A:4/8 W:4/8 O:14	A:4-8 W:4/8 O:18	A:8 W:8 O:22	A:8 W:8 O:24	A:8 W:8 O:23-27	A:4 W:4 O:5	A:1 W:1 O:7
Peak Energy Efficiency @INT8 (TOPS/W)	46.8	(1)42.5	18.1	16OCHs / PRE^3: 69.4 (4)64OCHs / PRE^3: 131.7	60.8	(2)76	(3)34.7
Full Precision?	YES	YES	YES	YES	NO	NO	NO

Fig. 6. Energy efficiency, MISS rate and STALL rate measurement results of ResNet-50 and comparison table.

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Chip Summary		
Technology	28nm	
Macro Capacity	Product-Cache: 144Kb Booth16 Multiplier: 22Kb	
Compute Circuit	Product-Cache + Addertree	
Supply Voltage	VDD_{AT}: 0.45-1.0 VDD_{MAIN}: 0.8-1.0	
Compute precision	A/W/O: 8/8/24	
Frequency	30-200MHz	
	16OCHs / PRE3	64OCHs / PRE3 *(Estimated)*
Input Channel	32	32
Output Channel	16	64
Macro Area	0.169 mm^2	0.571mm^2
Peak Throughput	0.205TOPS	0.819TOPS
(1)Energy Efficiency (Average Performance)	39.9TOPS/W	59.5TOPS/W
(2)Energy Efficiency (Peak Performance)	69.4TOPS/W	131.7TOPS/W

(1) *Measured with ResNet-50 under VDD_{MAIN}/VDD_{AT}=0.8/0.53V.*
(2) *Measured with the layer50 of ResNet-50 under VDD_{MAIN}/VDD_{AT}=0.8/0.53V.*

Die micro-graph and chip summary table.

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